Artificial Intelligence(23CS31T1)

Year: III B.Tech I Sem (2026-27) **Branch:CSE**

Assignment Questions

UNIT I

1. Discuss the foundations and history of Artificial Intelligence and its applications.

2. Analyze different types of intelligent agents and AI environments.

3. Illustrate the process of problem formulation in intelligent agents with a suitable example.

UNIT II

1. Compare Breadth First Search and Depth First Search algorithms with suitable examples.

2. Examine the working of A* and Hill Climbing search algorithms.

3. Demonstrate the Minimax algorithm with Alpha–Beta pruning in game playing.

UNIT III

1. Compare different knowledge representation techniques used in Artificial Intelligence.

2. Describe rule-based knowledge representation and deduction systems with suitable examples.

3. Evaluate Bayesian reasoning and Dempster–Shafer theory for reasoning under uncertainty.

UNIT IV

1. Explain First Order Logic and its inference mechanisms with suitable examples.

2. Differentiate forward chaining and backward chaining with suitable examples.

3. Discuss decision tree learning and reinforcement learning in Artificial Intelligence.

UNIT V

1. Illustrate the architecture and components of an expert system.

2. Justify the role of knowledge acquisition and heuristics in expert systems.

3. Assess the features and applications of MYCIN, DART, and XCON expert systems.